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Topic: Zomato Restaurants Analysis and Prediction

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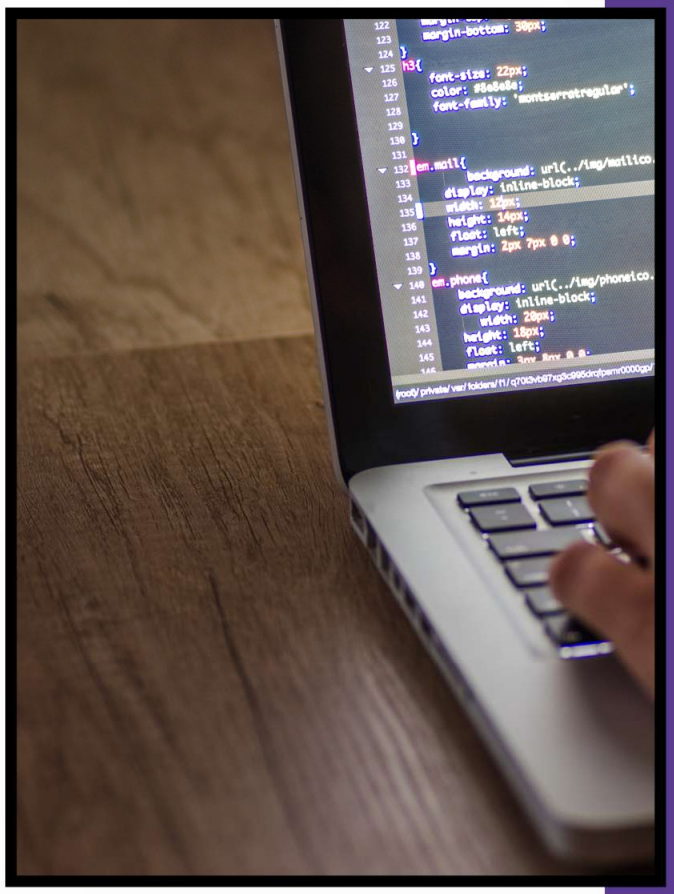
Introduction

- Zomato is a prominent meal delivery and restaurant discovery tool that offers consumers menus, reviews, and ratings for restaurants across the globe.
- Zomato's dataset provides valuable insights into consumer preferences, restaurant performance, and industry trends through the data generated by user interactions.
- Exploring the data, identifying relevant trends, and developing predictive models are the aims of the Zomato Restaurants Analysis and Prediction.
- These models are used to estimate a range of outcomes, including customer reviews, restaurant ratings, and even the success of a restaurant in a specific area.
- Due of its ability to forecast important elements influencing restaurant performance, this analysis might be useful to food connoisseurs, market analysts, and restaurant owners.



How we are solving

- **Project statement**: Zomato Restaurants Analysis and Prediction

- **Description:** We are using machine learning models to perform analytics on the dataset where we perform various things such as Data cleaning. I am performing analytics on the Zomato dataset to identify various trends in their dataset. First cleaning the dataset, it means removing missing values, and incorrect values, Data cleaning means fixing bad data in your data set. Bad data could be:
  Empty cells,
  Deleting redundant columns,
  Renaming the columns,
  Dropping duplicates,
  Cleaning individual columns,
  Removing the NaN values from the dataset,
  So that we can use a clean data set.

Algorithm and Flow Chart

1. Linear regression: It uses the relationship between the data points to draw a straight line through all of them. This line can be used to predict future values.

Python has methods for finding a relationship between data points and to draw a line of linear regression

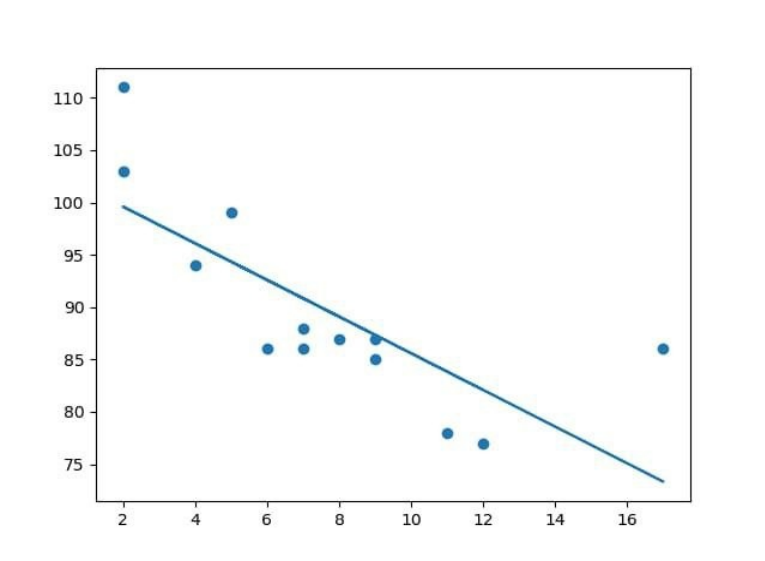
R for Relationship

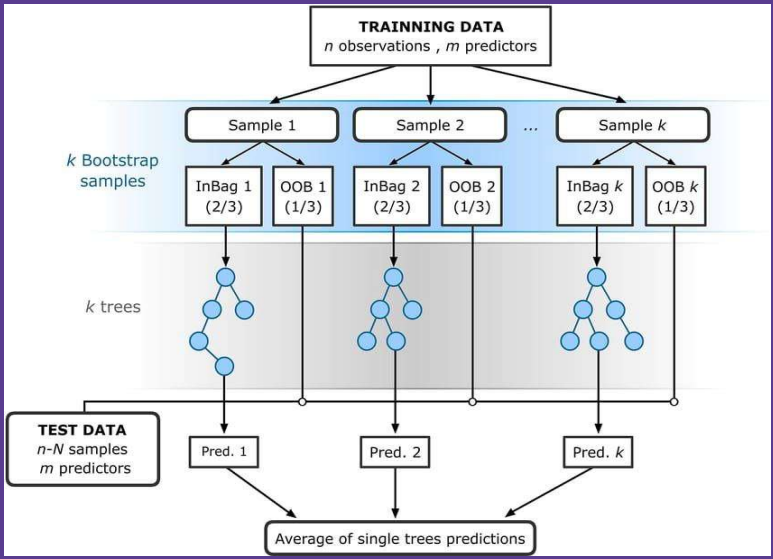
It is important to know how the relationship between the values of the x-axis and the values of the y-axis is, if there is no relationship the linear regression can not be used to predict anything.

This relationship - the coefficient of correlation - is called r.

The r value ranges from -1 to 1, where 0 means no relationship, and 1 (and -1) means 100% related.

Python and the Scipy module will compute this value for you, all you have to do is feed it with the x and y values.





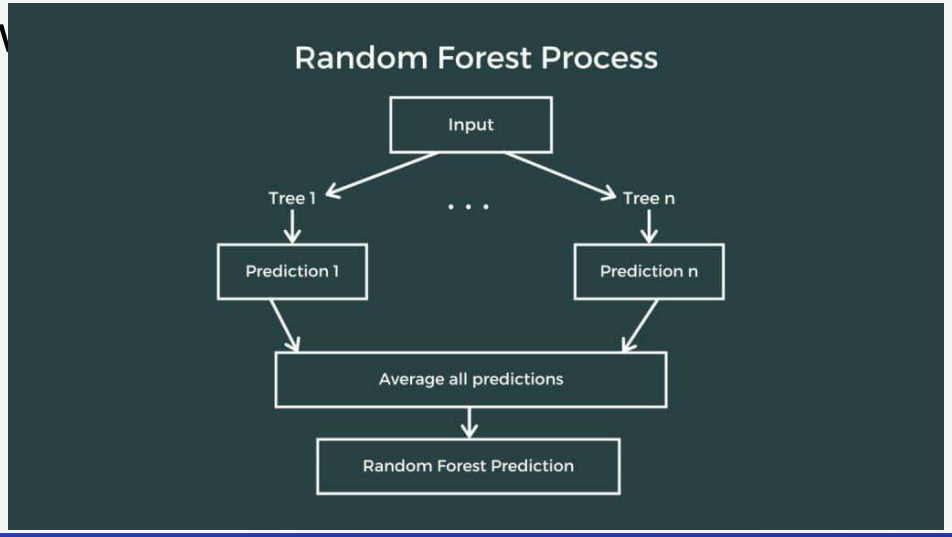
Algorithm:

2.Decision tree regression:

Decision Tree Regression is a supervised learning algorithm used for regression tasks. It works by recursively splitting the dataset into smaller and smaller subsets based on feature values. At each internal node of the tree, a decision is made by choosing a feature and a threshold to split the data such that the resulting child nodes are more homogeneous in terms of the target variable. The goal is to minimize the error in predicting continuous values

3. Random forest regression:

Random forest regression is a supervised learning algorithm and bagging technique that uses an ensemble learning method for regression in machine learning. The trees in random forests run in parallel, meaning there is no interaction between these trees



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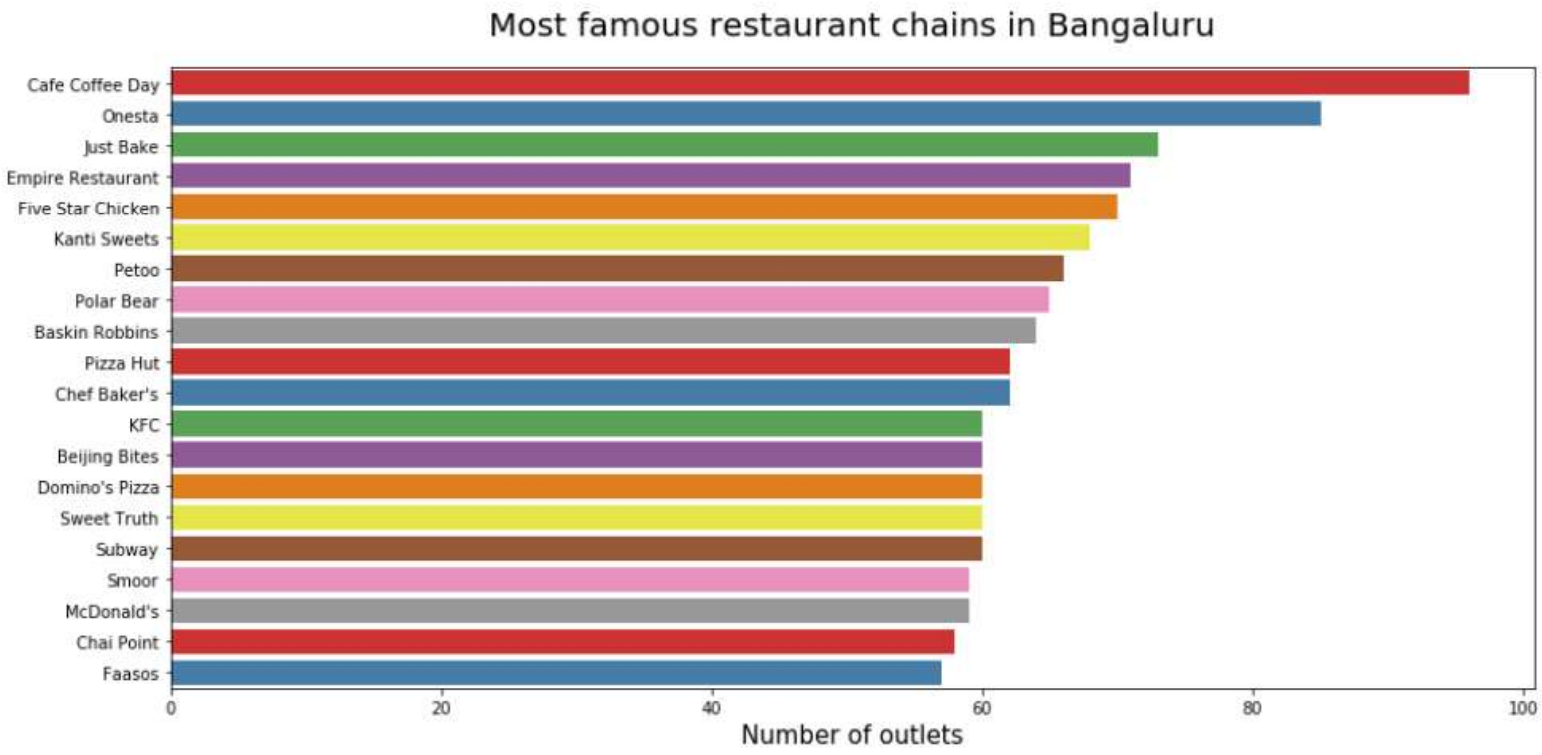
Exploratory Data Analysis refers to the critical process of performing initial investigations on data to discover patterns, to spot anomalies, to test hypotheses, and to check assumptions with the help of summary statistics and graphical representations" For this project we employed Python as an environment for all programing and execution needs. And used various Python Packages for our analysis such as:

1. Pandas
2. NumPy
3. Matplotlib
4. Seaborn
5.Scikit-learn

For the sake of a faster and more convenient understanding of our work this report only consists of derived results and graphs, Code to all I data processing, visualization, and results can be found in our Python Notebook here.

1. Top restaurant chains in Bengaluru



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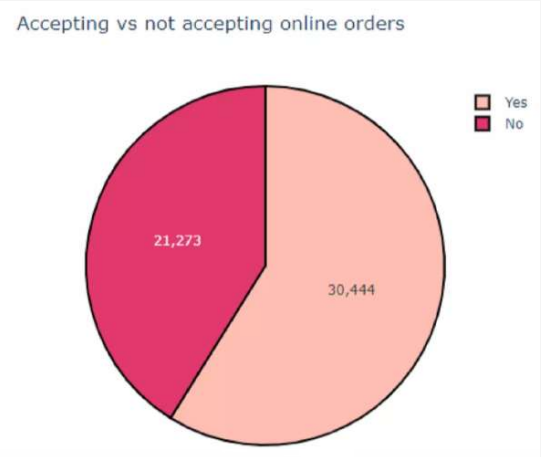
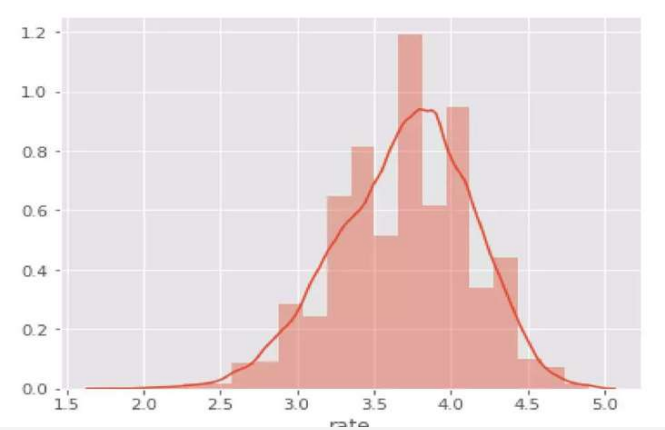
2. How many restaurants do not accept online orders?

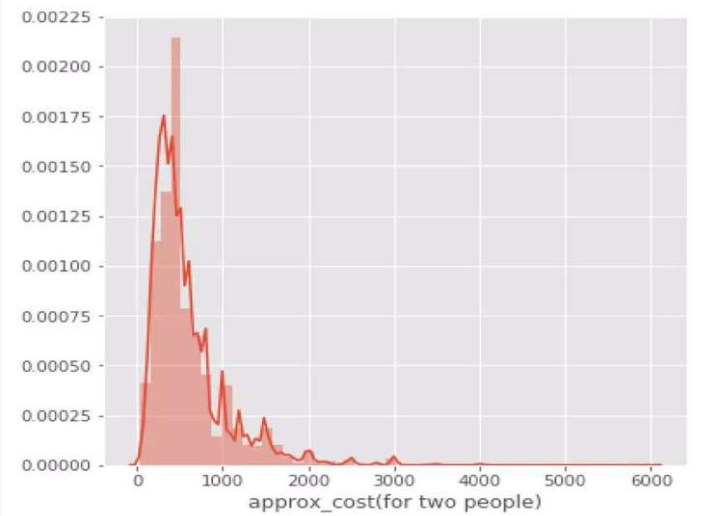


3. How many restaurants do not provide online table reservations?

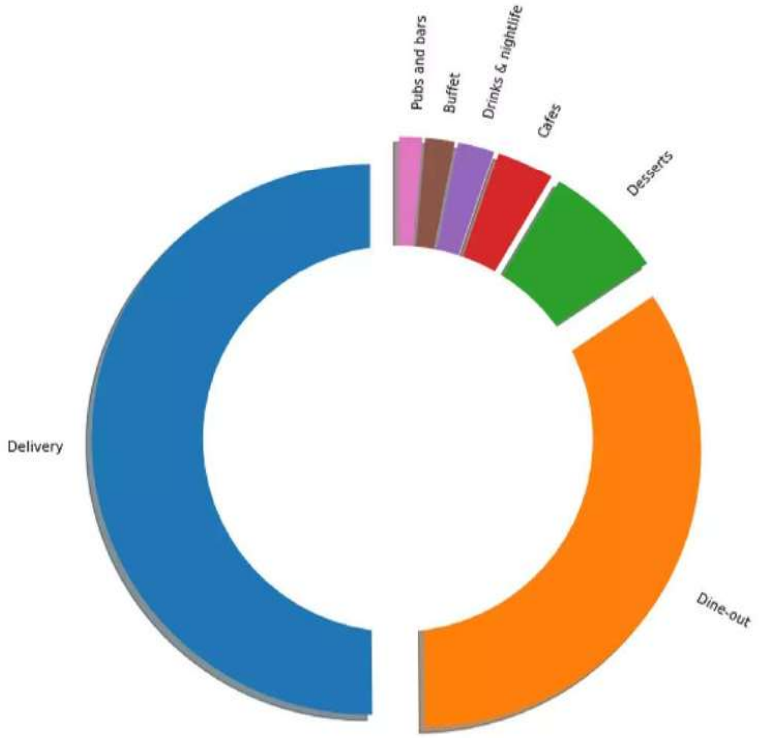
4. Distribution of Rating



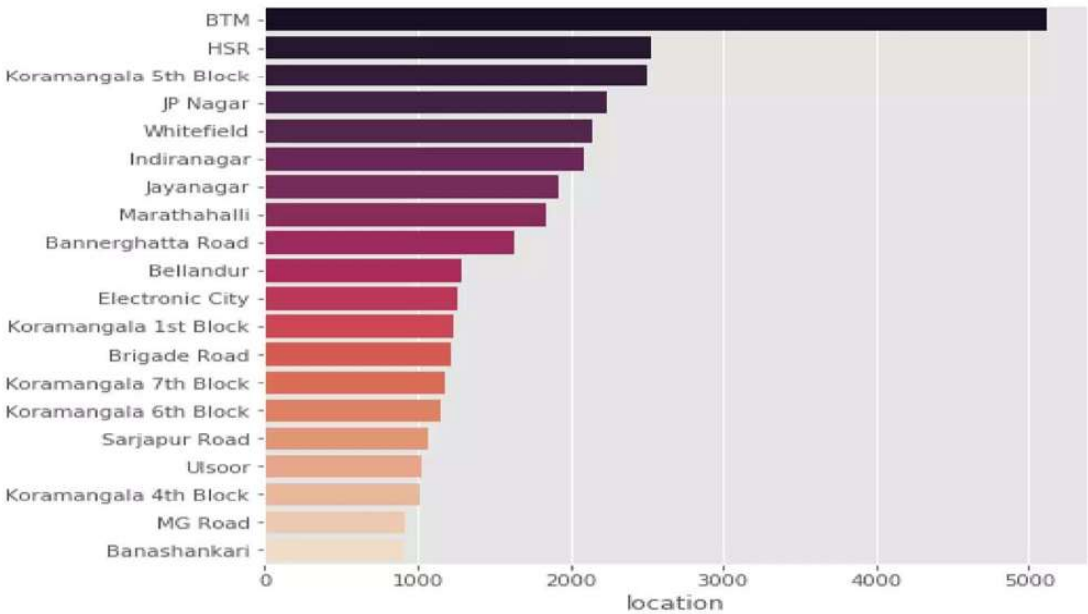
5. Distribution of cost for two people



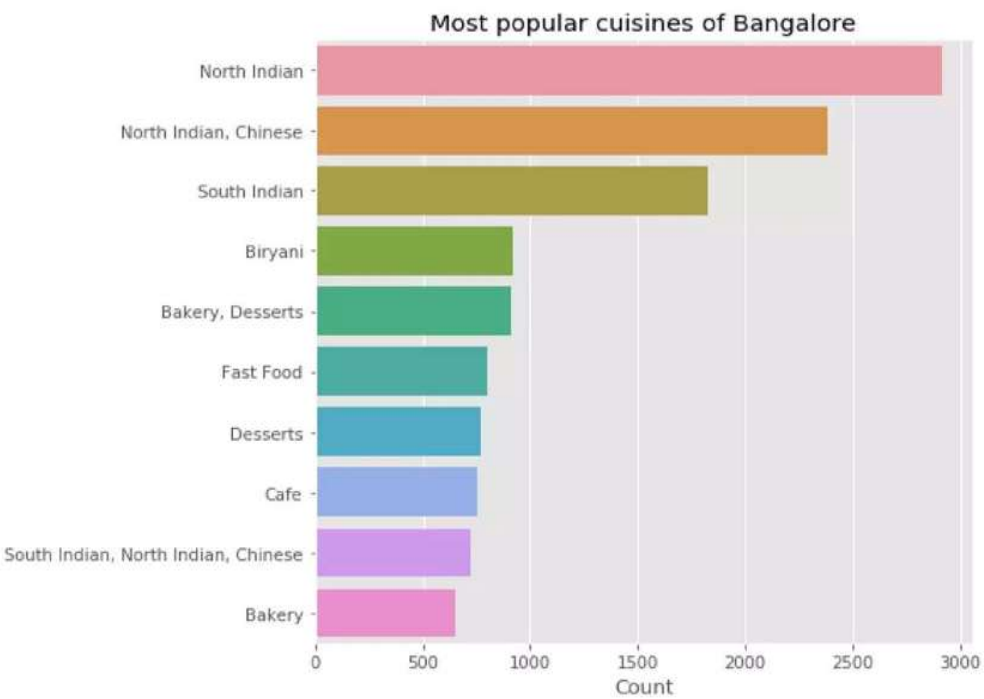
6. Distribution of type of service



7. Most foodie areas



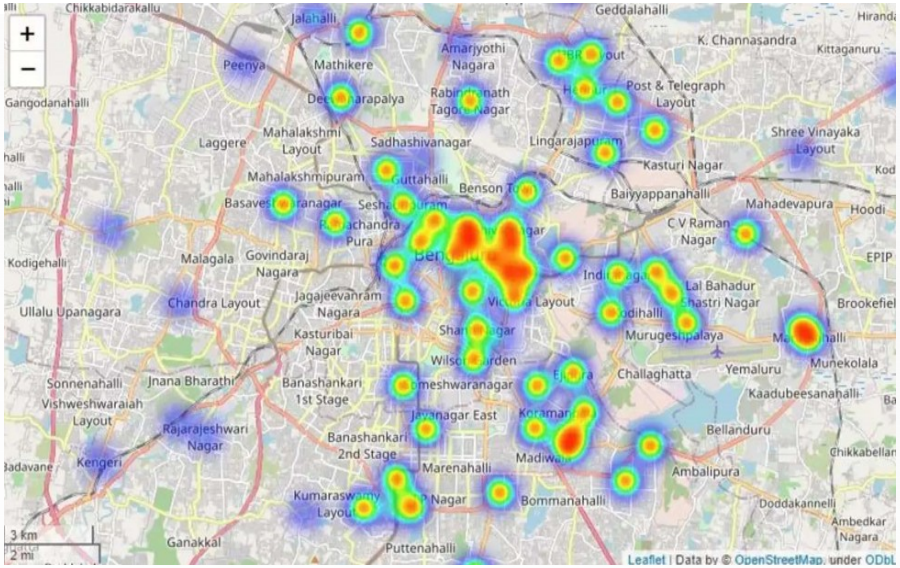
8. Popular cuisines in Bengaluru

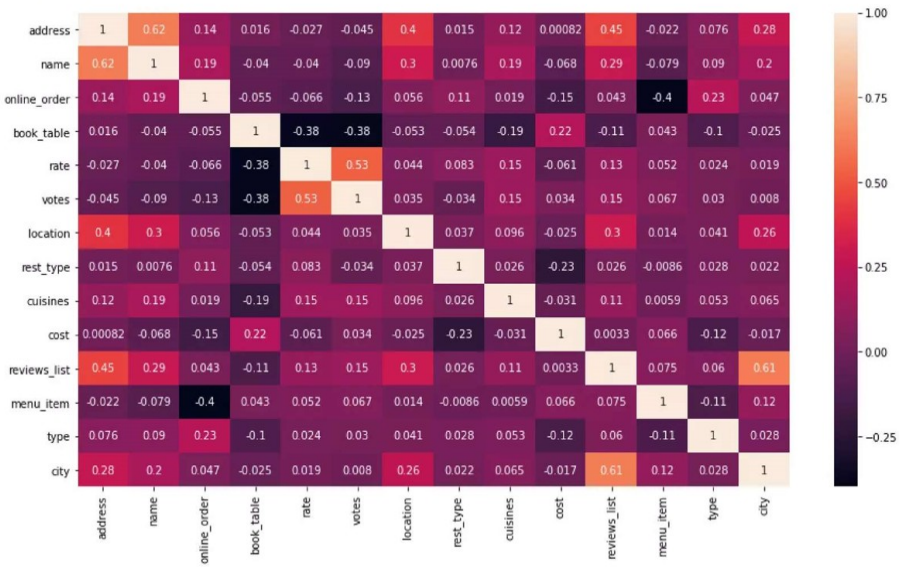


9. Top 15 highest-rated restaurants

	votes	rate
name		
Byg Brewski Brewing Company	16832	4.9
Toit	14956	4.7
Truffles	14726	4.7
AB's - Absolute Barbecues	12121	4.9
The Black Pearl	10550	4.8
Big Pitcher	9300	4.7
Onesta	9085	4.6
Arbor Brewing Company	8419	4.5
Empire Restaurant	8304	4.4
Prost Brew Pub	7871	4.5
Church Street Social	7584	4.3
Hoot	7330	4.2
Barbeque Nation	7270	4.8
Meghana Foods	7238	4.5
Flechazo	7154	4.9

10. Head map of restaurant count





11. Correlation between different variables

Conclusion:

- **The project successfully demonstrated the value of data analysis and machine learning in the restaurant industry by leveraging Zomato's extensive data. Through exploratory data analysis (EDA), we were able to gain valuable insights into customer preferences, popular cuisines, top-rated restaurants, and geographic distribution of restaurant ratings.**

References:

- **Source of Dataset**: https://www.kaggle.com/code/anshigupta01/bengaluru-zomato-restaurant-eda-and- prediction
- **Pandas Library :** https://pandas.pydata.org/docs/user_guide/index.html
- **Matplotlib Library:** https://matplotlib.org/3.3.1/users/index.html
- **seaborn** library: https://seaborn.pydata.org/tutorial.html

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